Emergent Specialization in Modular Recurrent Networks

Lesion Analysis of the FRANK Architecture

Izen Thornton

Independent Researcher

izen@cixate.com

Abstract

FRANK is a brain-inspired modular recurrent architecture presented in Part I [1]. Each module of FRANK is architecturally different (recurrence, memory, feedforward MLP), allowing gradient descent alone to determine what module to use depending on the task. I show this through targeted and global lesion studies across four algorithmic sequence tasks. Each task produces a unique pattern of component reliance; no routing signal, no task labels, no component-specific loss. Prior work found that models without routing affordances do not specialize [2]; FRANK produces a different outcome because its components have genuinely different inductive biases rather than being homogeneous experts differentiated by a router.

The brain modules (recurrent, stateful) are critical for every task. The reflex pathway (feedforward, stateless) is critical for running sum but dormant for copy and recall. Memory is critical for sum but nearly irrelevant for recall. The anomaly detector is dormant across all four tasks. None of this was prescribed; it emerges from the match between each component's architecture and each task's computational demands. I also observe an inverse relationship between structural damage robustness and length generalization: the most damage-tolerant model (GRU) generalizes worst at extreme scales, while the most fragile (FRANK) generalizes best.

1 Introduction

Modular architectures have produced good results in combinatorial generalization. Mixture-of-Experts [3, 4], RIMs [5], Neural Module Networks [6], and Routing Networks [7] all decompose computation across specialized modules. But every one of these uses explicit routing: attention-based selection, gating networks, or symbolic parsers to decide which module handles which input. Rosenbaum et al. [2] found that without routing, homogeneous modules do not specialize.

I test this claim using FRANK [1], a modular recurrent architecture whose components are architecturally heterogeneous rather than homogeneous. Using lesion studies inspired by neuroscience [8], I show that FRANK's five components develop distinct functional roles across four tasks through standard backpropagation. No routing needed.

2 Related Work

2.1 Modular Architectures and Specialization

MoE [3, 4] uses trainable gating. RIMs [5] use attention to select which modules receive input. Routing Networks [7] use RL-trained routers. In each case, an explicit mechanism assigns modules to inputs.

Work on emergent modularity has been mixed. Csordás et al. [9] found that standard networks specialize but do not reuse submodules. Mittal et al. [10] found that modular architecture alone is often not enough. Pfeiffer et al. [11] built a taxonomy of modular deep learning that assumes routing as a core component. FRANK (heterogeneous modules, zero routing) does not fit in that taxonomy. Filan et al. [12] showed that even monolithically trained networks are more modular than expected.

2.2 Lesion Methodology

Meyes et al. [8] formalized ablation methodology for neural networks. The Lottery Ticket Hypothesis [13] showed that specific weight subsets are critical for performance. The adversarial robustness-accuracy tradeoff has been studied by

Tsipras et al. [14], but structural damage robustness versus length generalization has not been explored.

3 Methods

3.1 Architecture

FRANK was developed in three stages (Modular → Modular+Memory → FRANK), each motivated by the previous design's limitations. All three are original to this work; full details are in Part I [1]. The brain-region labels in Figure 1 (cortex, hippocampus, brainstem, amygdala, basal ganglia) are loose functional analogies, not mechanistic claims; the components were inspired by those systems but do not simulate them. The three architectures share core components but differ in hidden dimensions and output heads, so this is a development trajectory, not a strict ablation. All use four tau-gated recurrent modules (simpler than GRUs; no gates, just a tau-damped recurrence) with learnable time constants (initialized at $\tau \in \{5, 15, 30, 50\}$, clamped to $[1, 100]$) and lateral connections applied after module updates. FRANK adds content-addressable active memory (64 learned key-value slots, queried using the previous timestep's brain state, injected into module 0), a feedforward reflex pathway (3-layer MLP, no recurrence), an anomaly detector (no direct connection to the output; trains only through gradient flow), and a veto mechanism. The brain produces a sigmoid inhibit signal (bias initialized at -1.0, so FRANK starts $\sim$73% reflex-dominated) that blends brain and reflex outputs. All components are active every timestep. No routing. Figure 1 shows the full architecture.

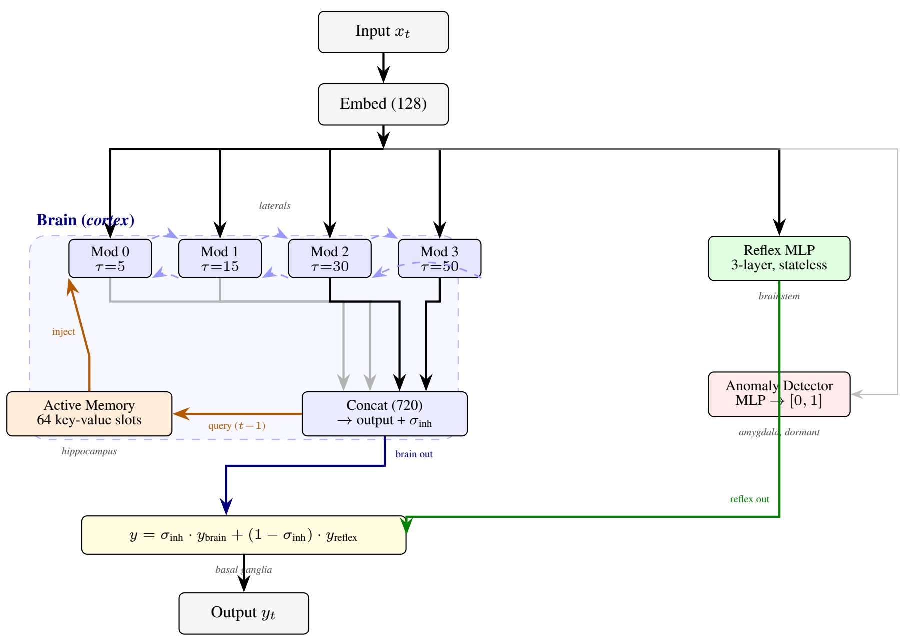

Figure 1: The FRANK architecture (reproduced from Part I). All components active every timestep; no routing.

3.2 Baselines

I compare against external baselines: GRU [15] ($\sim$500K), Transformer [16] ($\sim$481K), RIMs [5] ($\sim$500K), and FRANK-NoLat ($\sim$507K, laterals removed). Modular ($\sim$502K) uses four tau-gated modules with laterals and a 2-layer MLP output; no memory, reflex, or anomaly. Modular+Memory ($\sim$512K) adds active memory injected into module 0. FRANK ($\sim$507K) adds reflex, anomaly detector, and veto.

3.3 Lesion Protocol

Global lesion: Random weight zeroing applied uniformly across all parameters. Mean accuracy over 10 random damage patterns per level (0%, 10%, 20%, 30%, 50%).

Targeted lesion: Damage applied to one FRANK component at a time while all others stay intact. Damage levels from 0% to 70%, across all four tasks. Conducted on seed 42.

All model code, training configurations, and raw logs are available at https://github.com/izenthornton/frank-papers

4 Results

4.1 Targeted Lesion: Task-Dependent Activation

Table 1 shows what happens when each FRANK component takes 10% damage.

Table 1: FRANK accuracy at 10% targeted component damage (seed 42).

Component	Chain	Copy	Recall	Sum
Base (0%)	100.0%	95.5%	100.0%	96.7%
Brain 10%	58.1%	34.0%	83.0%	19.0%
Memory 10%	79.2%	77.8%	99.6%	13.0%
Reflex 10%	96.9%	95.5%	100.0%	20.5%
Anomaly 10%	100.0%	95.5%	100.0%	96.7%

Every task has a different sensitivity profile:

- **Chain:** Brain critical (58.1%), memory essential (79.2%), reflex degrades gracefully (96.9%), anomaly dormant.
- **Copy:** Brain critical (34.0%), memory supporting (77.8%), reflex dormant (95.5% = base), anomaly dormant.
- **Recall:** Brain moderately sensitive (83.0%), memory near-immune (99.6%), reflex dormant, anomaly dormant.
- **Sum:** Brain, memory, and reflex all collapse at once (13–20.5%). This is the only task with zero redundancy across three components.

4.2 Degradation Curves

Table 2 shows the chain task in more detail as damage increases.

Table 2: Chain task targeted lesion degradation (seed 42).

Component	0%	10%	20%	30%	50%	70%
Brain	100.0	58.1	32.4	21.2	17.9	17.3
Memory	100.0	79.2	30.9	21.6	17.5	17.3
Reflex	100.0	96.9	92.4	86.5	75.5	67.6
Anomaly	100.0	100.0	100.0	100.0	100.0	100.0

Three different patterns. The brain falls off a cliff: 100% to 58.1% at just 10% damage, then converges to a floor around 17%. Memory follows a similar curve but starts declining more gradually (79.2% at 10%) before hitting the same floor. The reflex declines smoothly and approximately linearly to 67.6% at 70%. It contributes, but it does not control the main computation.

4.3 Emergent Specialization Map

Table 3 puts the four-task picture together.

Table 3: Four-task component activation map. No two tasks share the same profile.

Component	Chain	Copy	Recall	Sum
Brain	CRITICAL	CRITICAL	CRITICAL	CRITICAL
Memory	Essential	Supporting	Near-immune	CRITICAL
Reflex	Graceful	Dormant	Dormant	CRITICAL
Anomaly	Dormant	Dormant	Dormant	Dormant

Four tasks, four different configurations. No routing told the model to do this. Rosenbaum et al. [2] found that homogeneous modules without routing do not specialize; in their setup, that held. FRANK shows that heterogeneity changes the outcome. Its modules have different computational structures (recurrence, memory, feedforward), and backpropagation allocates tasks to whichever component's inductive bias fits best.

4.4 Global Lesion: Cross-Architecture Robustness

Table 4 shows what happens when random weights are zeroed across each full architecture.

Table 4: Chain task global lesion (seed 42). Ranked by 10% damage retention.

Model	Base	10%	20%	30%	50%
GRU	100.0	**99.0**	70.5	36.4	18.4
RIMs	100.0	63.9	35.3	23.1	17.0
FRANK	100.0	49.3	18.1	13.9	11.8
Modular	100.0	46.3	25.4	20.1	14.2
FRANK-NL	100.0	40.2	18.5	13.6	10.4
Mod+Mem	99.9	39.1	22.4	17.6	13.5
Transformer	100.0	28.6	12.3	10.6	9.9

GRU retains 99.0% accuracy at 10% damage. FRANK retains 49.3%. The ordering is roughly the inverse of length generalization performance: GRU is the most damage-tolerant and generalizes worst at extreme scales; FRANK is the most fragile and generalizes best.

5 Discussion

5.1 Why Heterogeneity Works

Prior modular architectures use homogeneous experts and differentiate them through routing. FRANK's components are structurally different: recurrence for state maintenance, content-addressable memory for storage and retrieval, feedforward MLP for fast step-local computation, a statistical monitor for anomaly detection. Gradient descent allocates computation to whichever component fits; the reflex handles sum's step-local accumulation because it is feedforward; the brain handles chain's state tracking because it is recurrent; memory stores what recall needs stored. No one told the model to do this. The inductive biases did the work.

5.2 Robustness vs. Generalization

The inverse has a simple explanation. Monolithic architectures (GRU) spread information across a single large hidden state. Damaging part of it leaves partial representations intact, so the model is robust. But the same distributed representation drifts over long sequences as floating-point errors accumulate with no correction mechanism. Modular architectures (FRANK) put specific information in specific places. That makes them fragile (damage one component, lose that function) but also enables clean algorithmic behavior and error correction through lateral cross-checking. Robustness and generalization pull in opposite directions. Tsipras et al. [14] showed something similar for adversarial robustness vs. accuracy; the version I observe (structural damage vs. length generalization) is a different axis of the same kind of tradeoff.

5.3 The Dormant Anomaly Detector

The anomaly detector does nothing on these four tasks. Zero contribution at any damage level. This is expected: all four tasks present clean, in-distribution inputs at every test length. The anomaly detector was designed for distributional shift detection, and there is no distributional shift in these experiments. The dormancy is itself informative; FRANK does not force all components to participate. Specialization is emergent, not prescribed. Noisy inputs, regime changes, or adversarial perturbations might activate this component; that is future work.

5.4 Limitations

The main tables report targeted lesion results from seed 42. I collected targeted lesion data for all 10 seeds; preliminary analysis shows consistent profiles across seeds (brain always critical, anomaly always dormant, reflex graceful on chain), though the exact sensitivity magnitudes vary. Full multi-seed analysis is future work. Accuracy-based lesion is a coarse tool; a component labeled “dormant” may still contribute to gradient flow in ways I cannot measure. All four tasks are synthetic. Whether these specialization patterns hold on natural language or continuous control is unknown. The robustness-generalization inverse is an observation, not a proof.

6 Conclusion

Each module of FRANK is architecturally different, and gradient descent alone determines what module to use depending on the task. Four tasks produce four distinct activation profiles through standard backpropagation with no routing. Rosenbaum et al. [2] found that homogeneous modules without routing do not specialize; FRANK shows that architectural heterogeneity changes the outcome. Damage robustness and length generalization trade off against each other across architectures. Simplicity and specialization go together.

Part I [1] presents the FRANK architecture and its extreme length generalization results.

References

[1] Thornton, I. FRANK: Flexible Recurrent Active-memory Neural Kernel. arXiv preprint, 2026. [Part I.]

[2] Rosenbaum, C., Cases, I., Riemer, M. & Klinger, T. Routing Networks and the Challenges of Modular Computation. *arXiv:1904.12774*, 2019.

[3] Jacobs, R.A., Jordan, M.I., Nowlan, S.J. & Hinton, G.E. Adaptive Mixtures of Local Experts. *Neural Computation*, 3(1):79–87, 1991.

[4] Shazeer, N., Mirhoseini, A., Maziarz, K., et al. Outrageously Large Neural Networks: The Sparsely-Gated MoE Layer. *ICLR*, 2017.

[5] Goyal, A., Lamb, A., Hoffmann, J., et al. Recurrent Independent Mechanisms. *ICLR*, 2021.

[6] Andreas, J., Rohrbach, M., Darrell, T. & Klein, D. Neural Module Networks. *CVPR*, 2016.

[7] Rosenbaum, C., Klinger, T. & Riemer, M. Routing Networks: Adaptive Selection of Non-linear Functions. *ICLR*, 2018.

[8] Meyes, R., Lu, M., de Puiseau, C.W. & Meisen, T. Ablation Studies in Artificial Neural Networks. *arXiv:1901.08644*, 2019.

[9] Csordás, R., van Steenkiste, S. & Schmidhuber, J. Are Neural Nets Modular? *ICLR*, 2021.

[10] Mittal, S., Bengio, Y., Lajoie, G., et al. Is a Modular Architecture Enough? *NeurIPS*, 2022.

[11] Pfeiffer, J., Ruder, S., Vulic, I., et al. Modular Deep Learning. *TMLR*, 2023.

[12] Filan, D., Hod, S., Wild, C., Critch, A. & Russell, S. Neural Networks are Surprisingly Modular. *arXiv*, 2020.

[13] Frankle, J. & Carbin, M. The Lottery Ticket Hypothesis. *ICLR*, 2019.

[14] Tsipras, D., Santurkar, S., Engstrom, L., Turner, A. & Madry, A. Robustness May Be at Odds with Accuracy. *ICLR*, 2019.

[15] Cho, K., van Merriënboer, B., Gulcehre, C., et al. On the Properties of Neural Machine Translation. *SSST-8*, 2014.

[16] Vaswani, A., Shazeer, N., Parmar, N., et al. Attention Is All You Need. *NeurIPS*, 2017.